Supplementary material: comparison of p-OBOP formulations

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Abstract

This supplementary material accompanies the article *An optimization-based approach to ranking aggregation with weak-order outputs*. It compares assignment-based and representative-based formulations for the p-OBOP, examining the computational effects of redundant constraints, valid inequalities, and the relaxation of selected binary variables. Detailed running times and final optimality gaps are reported for different instances and prescribed numbers of buckets. A small example also shows that the optimal objective value need not be unimodal as a function of the number of buckets.

1 Comparison of p-OBOP models

This section analyzes the computational behavior of two families of equivalent mixed-integer programming formulations for the p-OBOP. On the one hand, we consider the formulations derived from model (2), which are based on assigning each item to exactly one of the p buckets. On the other hand, we study the formulations obtained from model (3), which rely on selecting representatives to define the bucket structure. Although all variants model the same optimization problem, they differ in the structural constraints they incorporate and in whether certain binary variables are relaxed to the continuous domain $[0, 1]$.

Throughout this supplement, $[n] = \{1, \dots, n\}$, and $f(\Pi, \boldsymbol{x})$ denotes the OBOP objective used in the accompanying article. All variants compared within each table have the same integer feasible outputs and aggregation objective.

Running times are reported in seconds, with runs reaching the time limit displayed as 3600. The relative optimality gap is computed as $100(z^{\mathrm{UB}} - z^{\mathrm{LB}})/z^{\mathrm{LB}}$, where z^{UB} denotes the best feasible objective value and $z^{\mathrm{LB}} > 0$ the final lower bound. Since the denominator is the lower bound, the gap may exceed 100%. A dash indicates that no feasible solution was found within the time limit. Times and gaps are rounded to two decimal places, so a displayed gap of 0.00% need not be exactly zero. Bold entries highlight selected best-performing configurations in each row, based on running time and final gap.

Before presenting the results, we recall that the base ordering inequalities used throughout the analysis are:

$$x_{rs} + x_{sr} \geq 1 \qquad \forall\, r, s \in [n] \,:\, r < s, \tag{1a}$$

$$x_{rs} + x_{st} \leq 1 + x_{rt} \qquad \forall\, r, s, t \in [n] \,:\, r \neq s,\ s \neq t,\ t \neq r. \tag{1b}$$

1.1 Variants of the assignment-based model

The formulation of interest is the following:

$$\begin{aligned}
\min \quad & f(\Pi, \boldsymbol{x}) && && \text{(2a)}\\
\text{s.t.} \quad & \sum_{u=1}^{p} y_{ru} = 1 && \forall\, r \in [n], && \text{(2b)}\\
& \sum_{r=1}^{n} y_{ru} \geq 1 && \forall\, u \in [p], && \text{(2c)}\\
& y_{ru} + y_{su} \leq x_{rs} + x_{sr} && \forall\, r < s,\ \forall\, u, && \text{(2d)}\\
& x_{sr} + \sum_{v=1}^{u} y_{rv} + \sum_{v=u+1}^{p} y_{sv} \leq 2 && \forall\, r \neq s,\ \forall\, u \in [p-1], && \text{(2e)}\\
& x_{rs} \in \{0, 1\} && \forall\, r \neq s, && \text{(2f)}\\
& y_{ru} \in \{0, 1\} && \forall\, r, u. && \text{(2g)}
\end{aligned}$$

Table 1 summarizes the computational results for the variants obtained by augmenting model (2) with the base inequalities (1a) and (1b), as well as by optionally relaxing the binary variables $\boldsymbol{x}$ to the continuous domain $[0, 1]$.

Instance	n	m	p	(2)		(2) + (1a)		(2) + (1b)		(2) + (1a) + (1b)		(2) + Relax $\boldsymbol{x}$		(2) + (1a) + Relax $\boldsymbol{x}$		(2) + (1b) + Relax $\boldsymbol{x}$		(2) + (1a) + (1b) + Relax $\boldsymbol{x}$	
				Time	Gap (%)	Time	Gap (%)	Time	Gap (%)	Time	Gap (%)	Time	Gap (%)	Time	Gap (%)	Time	Gap (%)	Time	Gap (%)
ED-10-02	51	9	2	**4.59**	0.00	10.01	0.00	1124.23	0.00	38.99	0.00	9.35	0.00	11.34	0.00	2680.83	0.00	3066.90	0.00
ED-10-03	54	10	2	**5.19**	0.00	7.83	0.00	1054.20	0.00	14.57	0.00	9.91	0.00	10.97	0.00	3202.51	0.00	3600	3.53
ED-10-11	50	11	2	**7.39**	0.00	9.85	0.00	1051.74	0.00	22.82	0.00	13.58	0.00	12.63	0.00	3486.87	0.00	3600	4.17
ED-10-14	62	15	2	**8.90**	0.00	9.85	0.00	3161.16	0.00	22.62	0.00	22.10	0.00	21.25	0.00	3600	117.32	3600	116.52
ED-10-15	52	14	2	**8.23**	0.00	10.66	0.00	1578.47	0.00	33.63	0.00	11.31	0.00	15.43	0.00	3600	14.23	3600	14.68
ED-15-20	122	4	2	25.31	0.00	**22.16**	0.00	3600	249.02	3600	25.44	2417.31	0.00	3600	4.97	3600	–	3600	–
ED-15-21	96	4	2	3.76	0.00	**2.38**	0.00	3600	278.45	3212.55	0.00	147.83	0.00	124.00	0.00	3600	–	3600	–
ED-15-22	112	4	2	6.06	0.00	**4.28**	0.00	3600	248.51	3600	44.75	325.79	0.00	976.62	0.00	3600	–	3600	–
ED-15-23	142	4	2	9.98	0.00	**6.73**	0.00	3600	285.59	3600	42.92	1398.02	0.00	1114.80	0.00	3600	–	3600	–
ED-15-24	91	4	2	3.16	0.00	**1.83**	0.00	3600	363.25	2169.20	0.00	63.17	0.00	142.08	0.00	3600	–	3600	–
ED-10-02	51	9	5	3600	269.43	3600	127.69	3600	84.64	**42.90**	0.00	3600	273.35	3600	198.51	3600	404.79	3600	229.71
ED-10-03	54	10	5	3600	208.06	3600	107.92	3600	175.98	**160.25**	0.00	3600	208.06	3600	182.37	3600	336.39	3600	368.91
ED-10-11	50	11	5	3600	198.60	3600	126.83	3600	70.60	**11.42**	0.00	3600	197.32	3600	225.44	3600	221.34	3600	272.88
ED-10-14	62	15	5	3600	286.95	3600	132.28	3600	87.52	3600	**0.15**	3600	266.93	3600	268.48	3600	457.14	3600	315.48
ED-10-15	52	14	5	3600	199.46	3600	90.75	3600	74.91	**3015.40**	0.00	3600	200.31	3600	187.88	3600	412.97	3600	279.67
ED-15-20	122	4	5	3600	763.11	3600	119.42	3600	84.80	3600	**0.59**	3600	763.11	3600	795.55	3600	–	3600	–
ED-15-21	96	4	5	3600	356.73	3600	26.36	3600	87.69	3600	**8.64**	3600	356.73	3600	337.02	3600	–	3600	213017.65
ED-15-22	112	4	5	3600	430.48	3600	23.40	3600	–	3600	**11.16**	3600	429.15	3600	394.22	3600	–	3600	–
ED-15-23	142	4	5	3600	741.91	3600	91.94	3600	88.04	3600	**16.56**	3600	741.91	3600	621.63	3600	–	3600	–
ED-15-24	91	4	5	3600	259.77	3600	28.21	3600	114.81	**1342.87**	0.00	3600	277.02	3600	280.64	3600	–	3600	–
ED-10-02	51	9	20	3600	806.42	3600	–	3600	**295.61**	3600	–	3600	2479.97	3600	2231.35	3600	352.49	3600	316.13
ED-10-03	54	10	20	3600	1026.68	3600	–	3600	390.53	3600	**20.05**	3600	3193.27	3600	2623.30	3600	322.45	3600	325.34
ED-10-11	50	11	20	3600	646.02	3600	642.73	3600	270.56	3600	**17.19**	3600	2799.74	3600	2917.96	3600	289.24	3600	226.44
ED-10-14	62	15	20	3600	482.54	3600	702.66	3600	248.15	3600	**19.68**	3600	3765.04	3600	5848.68	3600	301.58	3600	289.21
ED-10-15	52	14	20	3600	439.71	3600	478.08	3600	222.94	3600	**19.81**	3600	2926.52	3600	3961.12	3600	325.02	3600	264.03
ED-15-20	122	4	20	3600	100.37	3600	121.47	3600	282.43	3600	**2.69**	3600	691270.79	3600	395850.89	3600	–	3600	–
ED-15-21	96	4	20	3600	76.04	3600	137.67	3600	443.44	3600	**3.45**	3600	36400.84	3600	29764.27	3600	–	3600	–
ED-15-22	112	4	20	3600	90.03	3600	398.01	3600	275.86	3600	**2.85**	3600	105678.72	3600	152830.47	3600	–	3600	–
ED-15-23	142	4	20	3600	116.76	3600	125.10	3600	483.92	3600	**34.99**	3600	–	3600	–	3600	–	3600	–
ED-15-24	91	4	20	3600	108.03	3600	134.34	3600	171.75	3600	**2.82**	3600	34007.68	3600	7958.28	3600	–	3600	330337.37

Table 1: Comparison of model (2) variants for the p-OBOP. The term "Relax $\boldsymbol{x}$" denotes the relaxation of binary variables $\boldsymbol{x}$ to the continuous domain $[0, 1]$.

Table 1 shows that the computational effect of the additional constraints depends on the prescribed number of buckets. For $p = 2$, the assignment-based formulation without transitivity constraints and with binary $\boldsymbol{x}$ variables achieves the shortest running times, either with or without the before-or-after inequalities. For $p = 5$, including both families of inequalities while retaining binary $\boldsymbol{x}$ variables gives the strongest results in terms of running time or final gap. For $p = 20$, every variant reaches the time limit, although this same combination yields the smallest reported gap in nine of the ten instances. Relaxing $\boldsymbol{x}$ does not provide a consistent computational advantage. Thus, constraints that are redundant for the integer feasible set can nevertheless substantially affect solver performance.

1.2 Variants of the representative-based model

Recall that the model is the following:

$$\begin{aligned}
\min \quad & f(\Pi, \boldsymbol{x}) && && \text{(3a)}\\
\text{s.t.} \quad & \text{(1a)} - \text{(1b)} && && \text{(3b)}\\
& \sum_{r=1}^{n} \alpha_r = p && && \text{(3c)}\\
& \beta_{rs} \leq \alpha_s && \forall\, r, s \in [n] \;:\; r < s, && \text{(3d)}\\
& \sum_{s=r+1}^{n} \beta_{rs} + \alpha_r \;=\; 1 && \forall\, r \in [n], && \text{(3e)}\\
& x_{rs} \geq \beta_{rs} && \forall\, r, s \in [n] \;:\; r < s, && \text{(3f)}\\
& x_{sr} \geq \beta_{rs} && \forall\, r, s \in [n] \;:\; r < s, && \text{(3g)}\\
& x_{rs} + x_{sr} + \alpha_r \leq 2 && \forall\, r, s \in [n] \;:\; r < s, && \text{(3h)}\\
& x_{rs} \in \{0, 1\} && \forall\, r, s \in [n] \;:\; r \neq s, && \text{(3i)}\\
& \alpha_r \in \{0, 1\} && \forall\, r \in [n], && \text{(3j)}\\
& \beta_{rs} \in \{0, 1\} && \forall\, r, s \in [n] \;:\; r < s. && \text{(3k)}
\end{aligned}$$

We also considered the following valid inequality:

$$\beta_{rs} + \alpha_s \leq x_{rs} + x_{sr} \qquad \forall\, r < s. \tag{4}$$

Including inequality (4) and removing constraints (3f) and (3g) yields an alternative formulation of the problem that is equivalent to the original one.

Building on these results, a second set of experiments was conducted to evaluate the impact of specific inequalities on the performance of model (3). In this case, the objective is to determine which combinations or substitutions of inequalities lead to stronger formulations and faster convergence.

Instance	n	m	p	(3)		(3) − (3f) − (3g) + (4)		(3) + (4)	
				Time	Gap (%)	Time	Gap (%)	Time	Gap (%)
ED-10-02	51	9	2	995.41	0.00	1016.43	0.00	**493.93**	0.00
ED-10-03	54	10	2	**196.27**	0.00	869.89	0.00	366.32	0.00
ED-10-11	50	11	2	**10.56**	0.01	182.56	0.00	11.66	0.00
ED-10-14	62	15	2	**11.27**	0.00	21.31	0.00	12.07	0.00
ED-10-15	52	14	2	**781.46**	0.00	1172.56	0.00	889.97	0.00
ED-15-20	122	4	2	3600	**23.02**	3600	87.40	3600	23.02
ED-15-21	96	4	2	3600	32.54	3600	69.36	3600	**29.61**
ED-15-22	112	4	2	3600	**37.40**	3600	41.68	3600	37.41
ED-15-23	142	4	2	3600	–	3600	111.18	3600	**23.15**
ED-15-24	91	4	2	3600	**32.42**	3600	62.11	3600	61.43
ED-10-02	51	9	5	**51.86**	0.01	323.51	0.01	302.15	0.01
ED-10-03	54	10	5	**6.01**	0.00	47.08	0.00	34.54	0.00
ED-10-11	50	11	5	**5.93**	0.00	27.07	0.00	34.78	0.00
ED-10-14	62	15	5	**11.00**	0.00	71.60	0.00	66.62	0.00
ED-10-15	52	14	5	**7.70**	0.01	43.33	0.01	35.95	0.01
ED-15-20	122	4	5	3600	0.11	3600	3.40	3600	**0.07**
ED-15-21	96	4	5	3600	**1.26**	3600	2.75	3600	2.13
ED-15-22	112	4	5	3600	9.12	3600	98.29	3600	**7.91**
ED-15-23	142	4	5	3600	23.25	3600	46.12	3600	**22.27**
ED-15-24	91	4	5	**95.49**	0.00	3600	2.63	230.99	0.00
ED-10-02	51	9	20	3600	15.46	1015.58	0.00	**433.73**	0.00
ED-10-03	54	10	20	3600	4.59	860.85	0.00	**342.96**	0.00
ED-10-11	50	11	20	3600	15.78	181.26	0.00	**21.13**	0.00
ED-10-14	62	15	20	3600	0.60	21.28	0.00	**12.96**	0.00
ED-10-15	52	14	20	3600	10.19	1400.03	0.00	**883.35**	0.00
ED-15-20	122	4	20	3600	52.28	3600	87.40	3600	**23.02**
ED-15-21	96	4	20	**2145.80**	0.04	3600	69.36	3600	29.61
ED-15-22	112	4	20	3600	**37.40**	3600	41.68	3600	37.41
ED-15-23	142	4	20	3600	–	3600	111.18	3600	**23.15**
ED-15-24	91	4	20	3600	**32.42**	3600	62.11	3600	61.43

Table 2: Comparison of model (3) variants for the p-OBOP.

Table 2 shows that the additional valid inequality does not uniformly improve computational performance. For $p = 5$, the original representative-based formulation is the fastest on all five ED-10 instances. For $p = 20$, however, adding the valid inequality while retaining the original constraints enables all five ED-10 instances to finish with a displayed gap of 0.00% within the time limit, whereas the original formulation reaches that limit in every case. Results for the ED-15 instances are more mixed. These observations indicate that the benefit of strengthening the formulation depends on both the instance and the prescribed number of buckets.

2 Non-unimodality of the optimal p-OBOP value

Let $v(p)$ denote the optimal p-OBOP objective value. A natural question is whether every integer between two bucket counts attaining the unrestricted OBOP optimum also attains that optimum. The following example shows that this need not hold.

Example 1. *Consider the* OBOP *instance*

$$C = \frac{1}{100}\begin{pmatrix} 50 & 55 & 90 & 100 \\ 45 & 50 & 20 & 80 \\ 10 & 80 & 50 & 65 \\ 0 & 20 & 35 & 50 \end{pmatrix}.$$

The optimal p-OBOP *values are* $v(1) = 3.40$, $v(2) = 2.60$, $v(3) = 2.80$, *and* $v(4) = 2.60$, *as illustrated in Figure 2.*

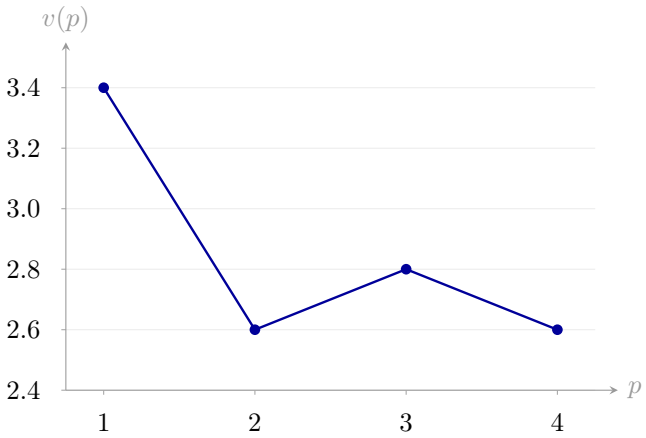


Figure 1: Optimal p-OBOP values for different numbers of buckets.

Figure 2: Optimal p-OBOP values. The dashed line marks the unrestricted OBOP optimum. Segments connect consecutive integer values of p for visual guidance.

Hence, the standard OBOP *has optimal solutions with* 2 *and* 4 *buckets, both with objective value* 2.60, *while the best solution with* $p = 3$ *has value* 2.80. *Therefore, the numbers of buckets associated with optimal* OBOP *solutions need not be consecutive, and the optimal value of the* p-OBOP *is not necessarily unimodal in* p.